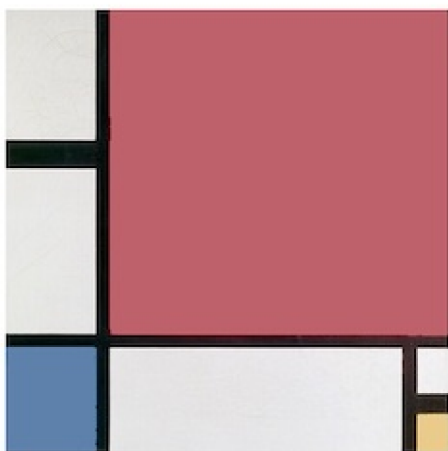


Visual Scene Graph

Baseline Experiment



Use a binding operation to encode the three colored features in this image, and to bind those three features to the overall scene. A memory query either completes the scene (replacement update rule) or predicts the missing elements (difference update rule).

```
<< "Circuits.m"
```

Scene completion

```
dataflow =  
  {  
    input[codec] [11] [],  
    auto["L"→True, "D"→0.95] [21] [11],  
    output[codec] [] [21],  
    _ → {2000, 10}  
  };
```

```
c = Circuit[ dataflow]; c[Graph, ImageSize→ 220]
```



Train

```
c[ {"01", "large", "red", "top", "right"}];  
c[ {"02", "medium", "blue", "bottom", "left"}];  
c[ {"03", "small", "yellow", "bottom", "right"}];  
c[ {"Scene", "01", "02", "03"}];
```

Analyze

```
c@ "Scene"
```

```
Sequence[{01, 02, 03, Scene}]
```

```
c@ "01"
```

```
Sequence[{01, large, red, top, right, 02, 03, Scene}]
```

```
c@ "red"
```

```
Sequence[{01, large, red, top, right}]
```

c@ "top"

Sequence[{01, large, red, top, right}]

c@ "bottom"

Sequence[{right, 02, medium, blue, bottom, left, 03, small, yellow}]

c@ {"blue", "medium"}

Sequence[{02, medium, blue, bottom, left}]

c@ "left"

Sequence[{02, medium, blue, bottom, left}]

c@ "right"

Sequence[{01, large, red, top, right, bottom, 03, small, yellow}]

c@ "small"

Sequence[{right, bottom, 03, small, yellow}]

c@ {"Scene", "blue"}

Sequence[{01, 02, 03, Scene}]